

Supervise the Belief, Not the Policy: Simulator-Supervised Belief for Partially-Observed Manipulation

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Abstract—How can simulation efficiently teach a visuomotor policy to gather information? Hidden physical properties, such as mass and friction, substantially affect how a robot can optimally manipulate objects. A privileged teacher directly accesses these properties, so its actions *generally do not demonstrate how to acquire missing information*, creating a mismatch for a partially-observing learner, which does not observe the privileged state during training or deployment. We instead use a simulator to simultaneously supervise a learned approximation to the belief state and train a policy conditioned on that approximation by reinforcement learning. Our method, Simulator-Supervised Belief Flow (SSB-Flow), learns a sampleable posterior over task-relevant physical quantities from interaction history, using high-dimensional image input alongside proprioceptive input. The policy consumes an encoding of posterior samples, allowing it to use estimated uncertainty when selecting actions. We compare SSB-Flow to three student-teacher approaches originally designed to use privileged information, as well as a strong recurrent RL baseline. Across four challenging manipulation tasks, SSB-Flow achieves the highest success rate on every task, averaging 93.1% versus 64.7% for the strongest alternative, an absolute gap of 28.4%. These results support supervising physical-state inference as an effective use of privileged information for partially observed manipulation. Code, results, and videos are anonymously available at <https://anonymous.4open.science/w/SSB-Flow/>

I. INTRODUCTION

Successful manipulation can depend on physical properties that are not directly observable, including mass, friction, and mass distribution. Objects with similar appearance may require different grasps, motions, or placements. A robot must infer these differences from the objects’ responses to interaction and use the accumulated evidence to select subsequent actions. This crucial information is often provided by simulators yet rarely directly observed with a robot’s sensors. Privileged-learning methods exploit this information in simulation through policy imitation or latent adaptation [1]–[3]. However, the best action under full observability is not necessarily the best action under partial observability. How can we exploit the privileged information that a simulator provides without leading the learner astray by the actions of a fully-observed teacher?

This mismatch, termed *an unidentifiable policy pair* [3], necessitates an alternative to attempting to match the actions of a privileged teacher. Our method uses a simulator to supervise a high-capacity conditional generative model to perform *state estimation*, and trains a policy by reinforcement learning on the estimator’s outputs. This approach is motivated by the belief-state formulation of partially observed control: under standard POMDP assumptions, the exact posterior over the Markov state summarizes the history sufficiently for op-

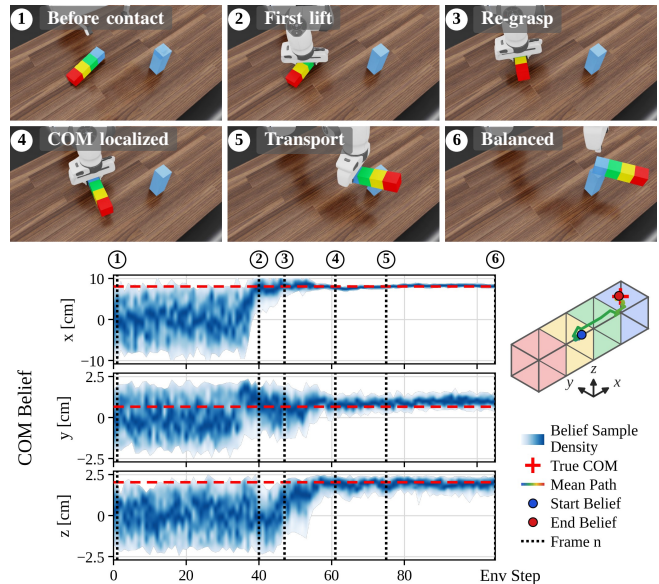


Fig. 1. SSB-Flow uses a deep flow-matching-based state estimator, supervised by a simulator, to efficiently inform a policy about uncertainty in the environment. Interactions help the state estimator to estimate physical properties, in this case, the unknown Center-of-Mass (COM) of the multicolored rod, which enables the policy to balance it on the blue pedestal. *Top*: A sequence of views during one episode. *Bottom-left*: The deep-state estimator’s belief samples for the COM. *Bottom-right*: The path of the estimated COM over the course of the episode. (1) Before contact, the COM belief is near-uniform over the rod’s extent. (2) The first grasp and lift collapses the belief along the rod’s long axis (x). (3) Re-grasping the upright rod begins to resolve the remaining lateral (y) and height (z) uncertainty. (4) The COM is now sufficiently localized, and the agent selects a grasp offset from the geometric center. (5) The agent transports the rod to the pedestal. (6) It releases the rod and successfully balances it.

timal control [4]. Although exact belief tracking is generally intractable, the formulation motivates learning a probabilistic state estimator together with a policy conditioned on its estimates. We introduce Simulator-Supervised Belief Flow (SSB-Flow; Fig. 2). A recurrent encoder summarizes camera images, proprioception, and previous actions; a conditional flow-matching model [5], [6] uses this summary to sample from a posterior over task-relevant physical quantities, which can remain multimodal while the history is ambiguous; and a permutation-invariant encoder [7] maps the samples to features for the actor.

We evaluate SSB-Flow on four manipulation tasks in which success depends on a hidden object property that can be revealed only through interaction. Our contributions are:

- 1) **Simulator-supervised probabilistic state estimation for manipulation.** We introduce SSB-Flow, which uses privileged simulator state to supervise a flow-matching state

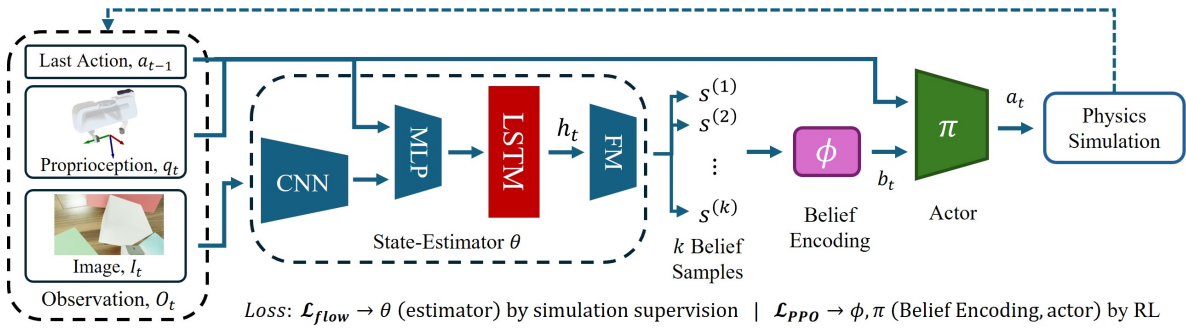


Fig. 2. SSB-Flow. **State-Estimator**: Given recurrent history h_t , the state estimator samples its belief from the history using Conditional Flow-Matching to estimate a compact state representation. **Belief Samples** ($s^{(1)}, s^{(2)}, \dots, s^{(n)}$): The samples reveal what the belief agrees on versus what remains uncertain, and when they split into several modes, they carry information that a mean and variance alone would discard. **Belief Encoding** ϕ : Deep Sets [7] invariant set encoding. **Policy-head** π : Both set encoder ϕ and actor head π are trained using PPO [8] on massively parallel environments.

estimator and trains an actor by reinforcement learning on encoded posterior samples. The actor receives no privileged input during training or deployment. Across all four manipulation tasks, SSB-Flow achieves the highest success rate on every task, averaging 93.1% versus 64.7% for the strongest alternative.

- 2) **Direct evaluation of physical-state estimates and uncertainty.** The estimator predicts physical quantities that can be evaluated against simulator ground truth independently of task success. We assess estimation accuracy, posterior dispersion and calibration, and the timing of uncertainty reduction relative to informative interactions. We also evaluate how the number of posterior samples and their encoding affect control performance.
- 3) **A manipulation benchmark for information acquisition.** We introduce four task variants involving hidden mass, friction, mass distribution, and an initially occluded color. The physical-property tasks require inference from interaction dynamics under the specified sensing setup; the color task provides a complementary test in which object reorientation makes the property directly observable. We compare all methods in simulation and evaluate hardware transfer on the occluded color task using a Franka FR3.

II. RELATED WORK

Distilling a Privileged Policy: RMA [1] trains a teacher on an encoding of the true physics parameters, then regresses an adaptation module onto that encoding from a window of proprioceptive history. RMA² [9] carries the recipe to manipulator arms, Visual Dexterity [10] and RotateIt [11] to in-hand reorientation, Learning by Cheating [12] to driving, and DextrAH-RGB [13] to grasping from RGB. In each, the teacher is trained first and is never modified, so everything the student receives arrives as an action target.

This works when the student can recover the teacher’s actions from its own observations. When it cannot, distillation of a frozen actor has no guarantee of successful task completion. The teacher optimizes over the true state and has no reason to restrict itself to behavior that is identifiable from the student’s history, so the student is regressed onto a target it cannot represent. Two methods address this by

making the teacher aware of the student. SITT [2] adds the predicted teacher-student action difference as a penalty in the teacher’s reward and aligns the two networks directly, which pushes the teacher toward behavior the student can imitate. A2D [3] makes the argument precise. It shows that asymmetric imitation recovers the optimal partially observing policy only for identifiable MDP-POMDP pairs, where the belief induced by the student’s history is enough to pin down the expert’s action. When the pair is unidentifiable, meaning that the belief induced by the student’s history is consistent with states at which the expert acts differently, the student can only learn the belief-average of the expert’s actions, and there is no reason for that average to be a good policy. A2D therefore adapts the teacher during training so that the expert is always one the student can follow.

Belief Representations and Learned State Estimators:

Recurrent latent-variable models learn a posterior jointly with the dynamics and condition the policy on it [14], [15]; FORBES [16] makes that posterior a normalizing flow so it can be multimodal. Meta-RL reaches the same construction as task inference [17], [18], the closest existing analogue to probe-then-commit. All are evaluated on gridworlds and low-dimensional control, where [19] show a tuned recurrent policy is already competitive, which is why PPO-LSTM [20] is our baseline in the same policy class.

Believer [21] models a belief over learned state features with a VAE and encodes posterior samples with a permutation-invariant network. BayesSim [22] likewise learns a posterior over physical parameters from simulated trajectories, which it uses to define the domain-randomization distribution during policy training. We retain this interface but target hidden physical parameters that must be inferred through contact, requiring the policy to select informative interactions for manipulation. Our conditional flow-matching formulation maps noise to physical-parameter samples conditioned on interaction history. In SSB-Flow, posterior samples condition the actor during execution. The estimator recomputes the posterior from the ongoing interaction history, allowing the policy to use estimated uncertainty when choosing information-gathering and task-directed actions.

III. PROBLEM FORMULATION

We model each task as a POMDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T, \omega, R, \gamma, \mu \rangle$, with transition kernel T , observation kernel ω , reward $R(s, a)$, discount $\gamma \in [0, 1)$, and initial state distribution μ . The agent observes o_t and selects a_t from its action–observation history $h_t = (o_0, a_0, \dots, a_{t-1}, o_t)$ to maximize $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)]$. The belief $\beta_t = p(s_t | h_t) \in \Delta(\mathcal{S})$ is a sufficient statistic for optimal control [4]. Writing τ for the Bayesian belief update, an optimal policy satisfies

$$\begin{aligned} \pi^*(\beta) &\in \arg \max_{a \in \mathcal{A}} Q^*(\beta, a), \\ Q^*(\beta, a) &= \mathbb{E}[R(s, a) + \gamma V^*(\tau(\beta, a, o'))], \end{aligned} \quad (1)$$

where $s \sim \beta$, $s' \sim T(\cdot | s, a)$, $o' \sim \omega(\cdot | s', a)$, and $V^*(\beta) = \max_a Q^*(\beta, a)$. Thus, optimal control can be expressed as state estimation followed by a memoryless policy in belief space, even when no observation-reactive policy is optimal.

We study manipulation tasks with state $s_t = (x_t^O, x_t^{\text{priv}})$, which we split into an observable part x_t^O , recoverable from passive observation (e.g., arm and object poses), and a privileged part x_t^{priv} of hidden task parameters that passive observation does not reveal. For mass, friction, and mass distribution, observations provide evidence through the object’s response to interaction. Actions therefore affect both task progress and the information available for subsequent decisions; information-gathering actions can be optimal despite providing little immediate reward.

This motivates learning a state estimator and a policy. Our approach is to learn a simulator-supervised state estimator using conditional flow matching, and condition a reinforcement-learned policy on samples from its estimated posterior. Specifically, we estimate a posterior over a task-specific simulator target s_t^b :

$$\begin{aligned} \hat{s}_t^{(i)} &\sim p_{\text{est}}(\cdot | h_t), \quad i = 1, \dots, K, \\ b_t &= E\left(\{\hat{s}_t^{(i)}\}_{i=1}^K\right), \\ a_t &\sim \pi(\cdot | q_t, a_{t-1}, b_t), \end{aligned} \quad (2)$$

where p_{est} approximates $p(s_t^b | h_t)$ and E is a learned permutation-invariant encoder. The simulator supplies supervision for the estimator, while the policy and set encoder are trained by reinforcement learning on task reward. Privileged state is available as an estimation target and critic input during training, but never as an actor input. The resulting representation approximates the information needed for optimal control. Although this representation is not guaranteed to be sufficient for optimal control, our experiments show that it supports effective information-gathering and task execution.

IV. A SIMULATOR-SUPERVISED BELIEF FLOW

Our method, SSB-Flow, implements the estimator–policy decomposition of Sec. III using a recurrent state estimator, a conditional flow-matching sampler, and a policy conditioned on an encoding of posterior samples (Fig. 2). Simulator state supervises the estimator, while task reward trains the policy. The actor receives estimated quantities during both training and deployment.

The deployable input is $o_t = (I_t, q_t, a_{t-1})$, comprising a wrist-camera image, proprioception, and the previous action respectively. The estimation target $s_t^b \in \mathbb{R}^d$ contains task-specific simulator quantities, including hidden physical parameters and dynamic state variables that require temporal inference. We denote the recurrent encoder and flow parameters by η and ξ , the set-encoder parameters by ζ , the actor parameters by θ , and the critic parameters by w .

A. Recurrent History Encoding

A recurrent encoder computes $h_t = f_\eta(o_t, H_{t-1})$. The encoder extracts image features, combines them with q_t and a_{t-1} , and updates an LSTM state that is reset at episode boundaries. The resulting h_t conditions the posterior sampler and provides history information to the critic. The actor accesses this history through the encoded posterior samples.

We train f_η exclusively through the state-estimation objective below. Simulator targets provide supervision at every timestep, allowing the recurrent representation to learn features predictive of the physical state. Policy and value gradients are blocked from the estimator, described in Sec. IV-D.

B. Conditional Flow-Matching State Estimation

The estimator represents $p_{\text{est}}(\cdot | h_t)$ through a conditional sampler $p_{\eta, \xi}(\cdot | h_t)$. Given h_t , a learned velocity field transforms Gaussian noise into samples of the target state. This construction enables multimodal estimates when several physical states are possible.

The components of s_t^b carry different physical units and scales, so we z-score each dimension before flow matching: $\bar{s}_t^b = (s_t^b - \mu)/\sigma$, with the subtraction and division taken per dimension and μ, σ the target’s mean and standard deviation, maintained as an exponential moving average over training iterations and held fixed within each update. Without it, the regression below is dominated by the largest-magnitude dimensions, and the Gaussian source of the flow is poorly matched to the target scale. For a noise sample $\epsilon \sim \mathcal{N}(0, I)$ and interpolation time $\alpha \sim \mathcal{U}[0, 1]$, define $y_\alpha = (1 - \alpha)\epsilon + \alpha \bar{s}_t^b$. Conditional flow matching [5], [6], [23] trains the velocity field u_ξ by minimizing

$$\mathcal{L}_{\text{flow}}(\eta, \xi) = \mathbb{E}\left[\|u_\xi(y_\alpha, \alpha, h_t) - (\bar{s}_t^b - \epsilon)\|_2^2\right], \quad (3)$$

where the expectation is over rollout timesteps, ϵ , and α . Gradients pass through h_t to the recurrent encoder. Training uses paired histories and simulator targets; it does not require multiple ground-truth states sampled conditionally on an identical history.

At action selection, we draw K independent Gaussian samples and integrate the velocity field with L Euler steps. For $i = 1, \dots, K$,

$$y_0^{(i)} \sim \mathcal{N}(0, I); \quad y_{\ell+1}^{(i)} = y_\ell^{(i)} + \frac{1}{L} u_\xi\left(y_\ell^{(i)}, \frac{\ell}{L}, h_t\right), \quad (4)$$

for $\ell = 0, \dots, L - 1$. Undoing the target standardization on $y_L^{(i)}$ yields $\hat{s}_t^{(i)}$. The samples are conditionally independent given h_t and the estimator parameters. All K samples are processed together at each integration step, requiring L sequential batched evaluations of the velocity field.

C. Belief Encoding

We encode the sample set using Deep Sets [7]:

$$b_t = E_{\zeta} \left(\{ \hat{s}_t^{(i)} \}_{i=1}^K \right) = \rho_{\zeta} \left(\frac{1}{K} \sum_{i=1}^K \phi_{\zeta}(\hat{s}_t^{(i)}) \right), \quad (5)$$

where ϕ_{ζ} and ρ_{ζ} are two-layer MLPs. The mean over samples makes b_t invariant to sample order.

Applying a learned nonlinear embedding before averaging allows b_t to encode distributional features beyond the sample mean, including dispersion and distinctions between modes. The set encoder is trained through the policy objective to extract features useful for action selection. Thus, b_t is a learned feature of posterior samples, distinct from the exact belief β_t defined in Sec. III.

D. Policy Learning

The actor is a Gaussian policy parameterized by an MLP: $a_t \sim \pi_{\theta}(\cdot | q_t, a_{t-1}, b_t)$. We train the actor and set encoder using the clipped PPO objective [8] with generalized advantage estimation [24]. No privileged expert supplies action targets; information-gathering behavior is learned through its effect on subsequent task return.

An asymmetric critic [25], [26] receives the recurrent history representation, proprioception, and privileged simulator parameters: $\hat{V}_t = V_w(\text{sg}[h_t], q_t, x_t^{\text{priv}})$, where $\text{sg}[\cdot]$ denotes stop-gradient. Including h_t allows the critic to account for information acquired through previous interactions. Different histories arriving at the same state may result in different beliefs, affecting future behavior and returns. This motivates a critic that also conditions on the agent’s history.

During rollout collection, posterior samples are generated without gradient tracking and stored with the transitions. PPO updates reuse these samples and recompute their encoding with the current set encoder. This holds the sample set fixed when comparing the current and behavior policies in the PPO importance ratio, while allowing the actor and set encoder to learn. The flow objective is evaluated on the collected observation sequences and simulator targets, with the recurrent encoder unrolled from stored recurrent states.

The resulting optimization uses disjoint parameter groups: $\mathcal{L}_{\text{flow}}$ updates (η, ξ) , while the PPO actor, value, and entropy losses update (θ, ζ, w) . Stored samples block policy gradients from entering the estimator, and $\text{sg}[h_t]$ blocks value gradients. The modules remain coupled through data collection: the policy determines which histories supervise the estimator, and the estimator determines the samples available for subsequent action selection.

At deployment, only the recurrent encoder, flow sampler, set encoder, and actor are required. The critic and simulator targets are discarded, and all actor inputs are computed from the robot’s observations and previous actions.

V. A HIDDEN-PROPERTY VISUOMOTOR MANIPULATION BENCHMARK

We evaluate on three tasks, four variants in total, all of which sit in the unidentifiable-process setting where the

TABLE I: The variables that S_t^b consist of in BALANCE (B), SORT tasks (S), HIDDEN-COLOR (H).

		dim	B	S	H
<i>Directly Observable Parameters x_t^O</i>					
p^{obj}	object position	3	✓	✓	✓
R^{obj}	object orientation	6	✓	✓	✓
v, ω	linear, angular velocity	3,3	✓	v	✓
$\Delta^{\text{bin}}_{1:3}$	object-to-bin offsets	6		✓	✓
$J^{\text{c}}_{1:3}$	contact impulse on object	3		✓	✓
p^{ped}	pedestal position	3	✓		
c^{world}	CoM in world frame	3	✓		
e^{align}	rod-pedestal alignment	2	✓		
<i>Hidden Task Parameters x_t^{priv}</i>					
$\log m$	mass	1	✓	*	✓
μ_s, μ_d	static, dynamic friction	2	✓	*	✓
c^{body}	COM offset, body frame	3	*		
$\log I_{\text{diag}}, I_{\text{off}}$	inertia tensor	3,3	✓		
κ	face color (one-hot)	3			*

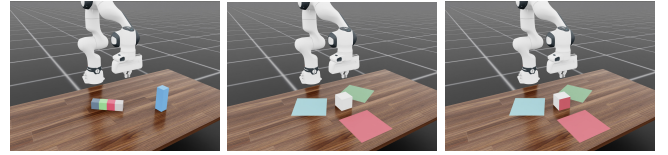


Fig. 3. Benchmark tasks. **Left:** BALANCE, place an unevenly weighted rod on a pedestal. **Middle:** SORT, bin by hidden mass or friction. **Right:** HIDDEN-COLOR, bin by the occluded downward face.

optimal MDP and POMDP policies differ [3]. We add one constraint to prior work: the information the agent needs must leave no trace in the environment’s persistent state, so passive observation does not reveal it and only the agent’s own interaction history does. For each task we define a belief target $s_t^b = (x_t^O, x_t^{\text{priv}})$. Table I lists both per task. Reward structure is documented on our project website.

BALANCE: Hidden Mass Distribution. The agent picks up an unevenly weighted rod and places it upright on a pedestal without dropping it and without it toppling after release. Where the center of mass sits is not visible and shows itself only in how the rod behaves once lifted. A release gate enforces the ordering: on release the gripper is forced open and withdraws, so a placement made before the mass distribution is resolved cannot be corrected.

SORT: Hidden Mass or Friction. The agent moves an object into one of three bins according to a hidden physical parameter, mass in SORT-MASS and dynamic friction in SORT-FRICTION. The object’s appearance is fixed across the parameter range, so the correct bin cannot be read off the image; the parameter appears only in the object’s response to contact, which the agent must generate before it can sort.

HIDDEN-COLOR: A Categorical Hidden Variable. The agent sorts a cube by the color of its downward face, occluded at episode start. The cube must be tilted or flipped until that face enters the camera’s view.

VI. EXPERIMENTS & RESULTS

We organize the evaluation around five research questions:

RQ1: How does SSB-Flow compare with recurrent reinforcement learning and privileged-policy transfer in

task success and training efficiency (Sec. VI-D)?

RQ2: How accurate and calibrated are the physical-state estimates, and how do they change during interaction (Sec. VI-E)?

RQ3: How does relative method performance vary across tasks with different information requirements (Sec. VI-F)?

RQ4: How do posterior sample count and integration steps affect performance (Sec. VI-G)?

RQ5: How well does SSB-Flow transfer to physical hardware (Sec. VI-H)?

A. Baselines

Privileged oracle. A PPO policy receives the full simulator state, including the hidden task parameters. Its performance provides a fully observed reference for each task, rather than a theoretical upper bound.

PPO-LSTM [8], [20]. A recurrent policy learns from task reward without an explicit physical-state estimation objective. It can condition its actions on the interaction history and learn information-gathering behavior.

RMA [1]. A privileged teacher is trained first, and an adaptation module learns to predict the teacher’s latent representation from the deployable history. We report the resulting student’s performance.

SITT [2]. The privileged teacher is trained with a penalty based on predicted teacher–student action disagreement, and the student learns from the resulting teacher.

A2D [3]. The expert is adapted to improve the return of the partially observing policy trained through asymmetric imitation. We report the deployed student’s performance.

B. Evaluation Metrics

Task success. A sorting episode succeeds when the object is placed in the bin selected by its hidden parameter or color. A BALANCE episode succeeds when the released rod remains balanced for 15 simulation steps. We compare success rates over evaluation episodes. For sorting, wrong-bin placements can additionally distinguish incorrect goal selection from other execution failures.

Training efficiency. We compare deployable-policy success against training environment transitions. Comparisons of total sample cost require consistent accounting of teacher pretraining and subsequent teacher/student training phases.

Estimation quality. We evaluate one SSB-Flow checkpoint per task against simulator ground truth on deterministic episodes. All estimator statistics are read at the commit step defined below, from 128 posterior samples (the actor pools $K=16$), and averaged over episodes. Error is the distance between the posterior mean and the true value, and dispersion is the posterior standard deviation; both are reported as a percentage of the quantity’s sampling range (mass: 0.2–3.0 kg; friction coef.: 0.2–1.0; COM: the range of true body-frame COMs per axis, averaged over axes). Calibration is the fraction of true values falling inside the posterior’s central 90% interval, so a calibrated estimator scores 0.90 and a smaller value means overconfidence. HIDDEN-COLOR is

categorical: each sample decodes to a color class, error is the misclassification rate, and calibration is the gap between predicted confidence and observed accuracy, ideally 0.

Interaction timing. We locate two events per episode and report both as median seconds from first contact with the object. The belief settles once its dispersion has fallen to a quarter of its value at the start of the episode (color: once the top-class probability first reaches 0.9). The agent commits once it stops gathering information and executes task-directed actions: in sorting, the first step from which the object only ever moves toward the bin it ends in; in BALANCE, the first step from which the agent holds the rod and moves it steadily toward the pedestal. Episodes that never settle are excluded from settle times. Error and calibration measures estimate correctness, whereas settling measures sample concentration. Comparing the two events shows whether uncertainty is resolved before the decision that depends on it.

C. Experimental Setup

We train in Isaac Lab with 4096 parallel environments for 500M transitions (4096×16 per PPO update, training environments only), with 3 seeds per method and task. Although large by RL standards, this budget is cheap under GPU-parallel simulation: one SSB-Flow run takes ~ 13 h on a single RTX 5090. The deployable input $o_t = (I_t, q_t, a_{t-1})$ is a 64^2 wrist-camera RGB image, proprioception, and the previous action. SITT and A2D train teacher and student on shared rollouts, so their counts include both; RMA trains sequentially, so its student curve is shifted by its teacher’s transitions to convergence. All methods share one PPO implementation, and the partially observing baselines use the recurrent encoder of Sec. IV-A. SSB-Flow uses privileged state only as an estimation target and critic input, never as an actor input; transfer baselines use it only for teacher training. Deployable policies receive only their permitted observations; the privileged oracle is evaluated with full state. Evaluation runs on 64 extra environments, excluded from training and the transition count, using the current policy’s mean action; success is averaged over the latest 512 episodes. SSB-Flow draws $K=16$ belief samples per step in training and evaluation. For distillation baselines, only the student is evaluated.

D. Simulation Performance

Table II addresses the task-performance component of RQ1. SSB-Flow achieves the highest success among deployable methods on all four tasks. On BALANCE, SORT-MASS, SORT-FRICTION, and HIDDEN-COLOR, it improves on PPO-LSTM by 39.9, 25.5, 19.5, and 15.2 percentage points, and on the best distillation baseline by 35.7, 7.0, 38.5, and 65.0. It comes within 7.2 points of the privileged oracle on the sorting tasks and HIDDEN-COLOR, and reaches 83.4% on BALANCE against the oracle’s 95.9%. These comparisons evaluate the complete SSB-Flow method, including its estimation objective, architecture, and posterior-sample interface. The training-efficiency component of RQ1

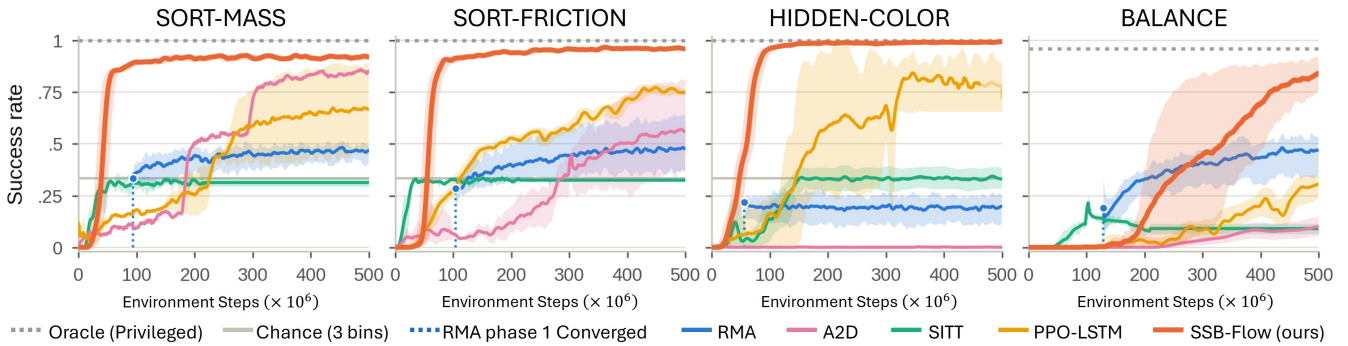


Fig. 4. Deployable-policy success rate against training environment steps on SORT-MASS, SORT-FRICTION, HIDDEN-COLOR, and BALANCE. Lines are the mean over 3 runs per method, smoothed over 5M steps. Bands span the minimum to maximum across runs. Success is measured by online evaluation (Sec. VI-C). Grey dotted: best Oracle success; grey solid: chance (three bins). RMA’s student starts where its teacher converged (blue dotted).

TABLE II: Simulation success rates: the peak of each method’s curve in Fig. 4 within 500M training steps, averaging its 3 runs and smoothing over 5M steps. We report online evaluation of the deployable policy acting on its mean action. Bold indicates the best deployable method; the privileged oracle receives full-state inputs.

Method	BALANCE	SORT MASS	SORT FRICTION	HIDDEN COLOR	Average (δ vs. Ours)
Privileged oracle	95.9%	100.0%	100.0%	100.0%	99.0% (+5.9%)
RMA	47.7%	48.2%	48.6%	20.5%	41.3% (−51.8%)
SITT	17.9%	32.7%	33.4%	34.4%	29.6% (−63.5%)
A2D	9.8%	85.8%	56.8%	0.3%	38.2% (−54.9%)
PPO-LSTM	30.2%	67.3%	77.2%	84.2%	64.7% (−28.4%)
SSB-Flow (Ours)	83.4%	92.8%	96.7%	99.4%	93.1% (0%)

is evaluated through the learning curves in Fig. 4. Final success rates alone do not determine how many transitions each method requires to reach a given performance level.

E. State Estimation During Interaction

RQ2 evaluates physical-state estimation independently of task success. We compare estimator outputs with simulator ground truth and examine how accuracy and uncertainty change as the robot interacts with the object. Estimates before informative interaction provide a reference for assessing the information acquired during manipulation.

For SORT-MASS and SORT-FRICTION, the analysis concerns the parameter that determines the correct bin. For BALANCE, it concerns the physical quantities used to characterize the rod and its placement. HIDDEN-COLOR provides a complementary case: the relevant variable becomes directly observable after the object is reoriented.

Table III and Fig. 5 report aggregate measurements and one illustrative episode, respectively. The belief almost never settles before the robot has touched the object (0% of BALANCE episodes, under 0.3% in sorting), and in BALANCE three quarters of its total narrowing over the episode is complete before the agent begins transporting the rod, thus showing information comes from interaction, and the policy acts once it has it.

F. Performance Across Information Requirements

RQ3 examines how relative performance varies across the benchmark. SSB-Flow’s margin over PPO-LSTM is largest

TABLE III: Estimator quality at the commit step. Metrics are defined in Sec. VI-B. Error and Disp. are percentages of each quantity’s sampling range. Cal. is ideal at 0.90, and for color is the confidence–accuracy gap, ideal at 0. Settle and Commit are median seconds from first contact. Beliefs are accurate to within 10% of the parameter range at commit step, mildly overconfident, and most of their narrowing precedes the commit.

Task	Quantity	Error	Disp.	Cal.	Settle	Commit
BALANCE	COM	10.0	9.3	0.76	0.40	0.55
SORT-MASS	Mass	4.9	3.7	0.70	0.40	1.30
SORT-FRICTION	μ_d	2.6	2.1	0.70	1.05	1.15
HIDDEN-COLOR	Color	0.8	–	0.007	0.25	0.70

on BALANCE (53.2 points), the task with the largest estimation target (Table I), and smallest on HIDDEN-COLOR (15.2). One possible explanation is that direct physical-state supervision is particularly useful when several coupled quantities must be inferred from interaction. We hypothesize that the estimator helps learn these quantities by providing a supervised target at each timestep. The task comparison does not, however, isolate estimation difficulty from differences in control, reward structure, and exploration.

HIDDEN-COLOR requires an action that changes visibility rather than inference from a physical response. SSB-Flow’s 15.2-point improvement over PPO-LSTM on this task shows that its usefulness is not confined to estimating mass or friction. A possible explanation is that the estimation objective helps the recurrent representation retain task-relevant evidence after it becomes visible.

The distillation baselines exhibit different patterns. RMA’s teachers reach 94–100% success (one SORT-FRICTION teacher: 67%), but its students reach only 20.5–48.6%: the adapter never recovers the teacher’s latent. It is weakest on HIDDEN-COLOR, where the decisive color is exposed only briefly, so a point estimate that misses that moment cannot recover, whereas mass and friction are re-expressed at every contact. On SORT-MASS, 33–44% of student episodes end in the wrong bin, indicating estimation rather than execution errors.

SITT’s students stay at chance on both sorting tasks and HIDDEN-COLOR (32.7–34.4%), although their teachers succeed in 99.5% of sorting episodes. On SORT-MASS

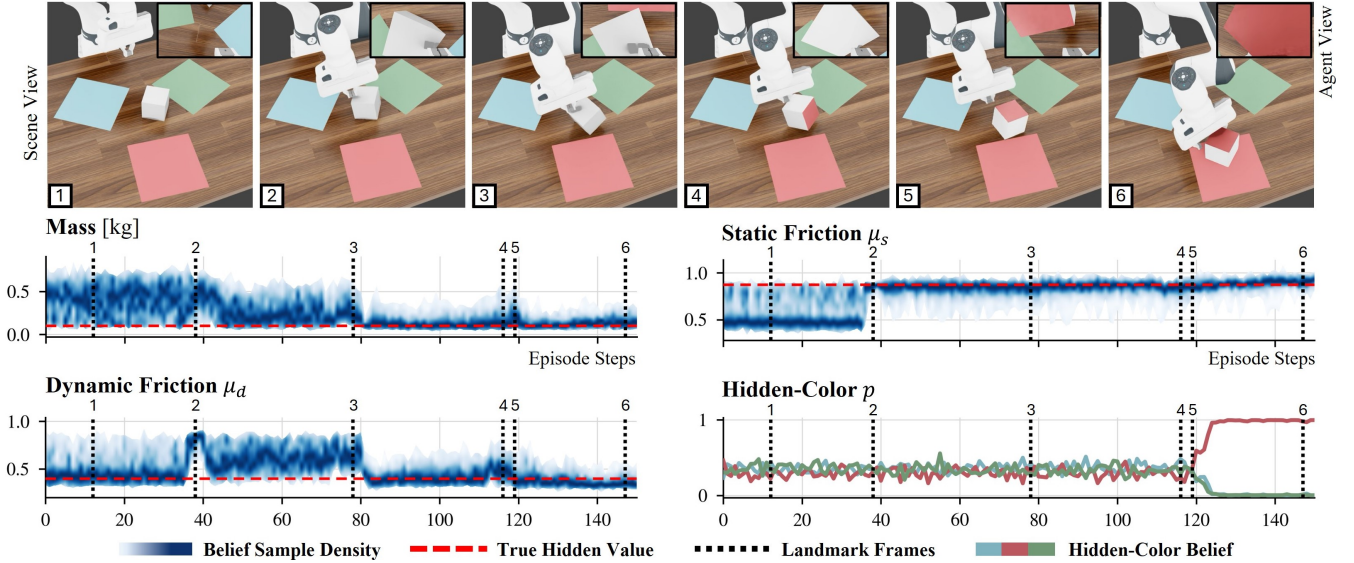


Fig. 5. Physical-state estimates during interaction on the HIDDEN-COLOR task. [1] Before any contact, mass and hidden-color beliefs are near-uniform, and the friction beliefs sit at their prior — notably underestimating the true static friction. [2] The first contact and failed flip sharpen the static-friction belief to its true value; the mass belief narrows but remains overestimated, while dynamic friction temporarily overshoots. [3] A second failed flip collapses both the mass and dynamic-friction beliefs onto their true values. [4] With the physical parameters now accurately estimated, the third interaction successfully flips the object. [5] The first frame in which the hidden color is observed by the agent. [6] The object is correctly sorted.

the SITT students place the object in 93–96% of episodes but choose the wrong bin in 62–66%: they reproduce the teacher’s execution but not its bin choice. On BALANCE, the teacher reaches 78% and the student 17.9% at its best. This is consistent with SITT and RMA relying on the supervision from a privileged teacher that does not exhibit the information-gathering behavior required to solve the tasks.

A2D varies most across tasks: 85.8% and 56.8% on the sorting tasks, but 9.8% on BALANCE and 0.3% on HIDDEN-COLOR. Its online-evaluated SORT-FRICTION run reaches 56.8% with 27% wrong-bin placements. On HIDDEN-COLOR it reaches any bin in 0.3% of episodes, an execution or exploration failure rather than a wrong choice. Success alone does not establish which of these is responsible. We note that with appropriate hyperparameter tuning A2D may approach in the limit similar performance as SSB-Flow, but within the given step budget, A2D is highly sample inefficient compared to SSB-Flow.

G. Ablations and Sensitivity

RQ4 varies the posterior sample count K and integration steps L relative to a reference with $K=16$ and $L=6$. Sensitivity rows evaluate our best BALANCE checkpoint (3.3B transitions) with a different K or L ; scratch rows change one component and train from scratch for 500M transitions (one seed each) against a reference trained identically shown in Table IV. The trained policy is insensitive to K (at most 1.5 points) but loses 7.9 points with a single integration step, which lowers COM interval coverage from 0.76 to 0.31 by pulling samples toward the posterior mean. Trained from scratch with $K=1$, the policy never learned the task. Trained with $L=1$, it matches its reference (−0.1), so the cost of $L=1$ at test time comes from the mismatch with training rather

TABLE IV: Ablations and sensitivity on BALANCE (RQ4). Sensitivity: the best checkpoint (3.3B transitions) evaluated with a different K or L . Scratch: one component changed and trained from scratch for 500M transitions (one seed each). Δ : percentage points from the reference.

Mode	Configuration	Success	Δ
Reference	$K=16, L=6$ (3.3B)	96.8	0.0
Sensitivity	$K=1$	95.6	−1.2
	$K=4$	95.3	−1.5
	$K=32$	96.2	−0.6
	$L=1$	88.9	−7.9
	$L=12$	97.0	+0.2
Reference	$K=16, L=6$ (500M)	78.5	0.0
Scratch	$K=1$	0.0	−78.5
	$L=1$	78.4	−0.1

than from one step itself, showing that SSB-Flow may not need more integration steps.

H. Real-Robot Evaluation

RQ5 concerns transfer to physical hardware. We deploy on a Franka FR3 the HIDDEN-COLOR task with an 8 cm white cube with one colored face, placed colored face down so that every trial requires a flip (Fig. 6). The policy observes a 54×96 RGB image from a wrist-mounted ZED Mini, proprioception, and its previous action at 20 Hz, and commands Cartesian end-effector displacements tracked by an impedance controller that simulation reproduces. Training randomizes backgrounds, lighting, image photometrics, camera extrinsics, robot base pose, initial joint configuration, end-effector velocity scale, and object mass and friction. A trial succeeds when the cube is placed in the bin matching its color. SSB-Flow succeeds in 24 of 30 trials. PPO-LSTM, trained under the same randomization, does not learn the task

TABLE V: Hardware evaluation with a Franka FR3 (success / trials).

	PPO-LSTM	SSB-Flow
HIDDEN-COLOR	0/30	24/30



Fig. 6. SSB-Flow execution on the Franka FR3 performing the HIDDEN-COLOR task. The policy first tips the cube over, then checks the new side revealed and confirms the color (red) before sorting it to the correct (red) bin.

in simulation (0.1% success) and succeeds in 0 of 30. SSB-Flow’s failures arise from small discrepancies between commanded and executed motions, relative to simulation, which lead to unseen states. Please check the video supplementary material for demonstration videos.

VII. LIMITATIONS

As the estimator and the actor are separate modules with disjoint parameters, a converged estimator can in principle be frozen and reused while a fresh actor trains on top. We do not test this. Every result trains both from scratch, and sample cost of pretraining is unmeasured. Second, the estimator is task-specific. Its target s_t^b is defined per task by hand, so two tasks share an estimator only if they share the same target. Covering a family of tasks with one estimator needs a target that generalizes across them, which we leave to future work.

VIII. CONCLUSIONS

We study manipulation tasks where task-relevant physical state is hidden and must be inferred through interaction. In this setting, privileged teacher actions alone provide an incomplete training signal because they do not teach the student information-gathering behavior. Our method, SSB-Flow, learns a sampleable posterior over task-relevant physical quantities from interaction history. The policy conditions its actions on an encoding of samples from the posterior, allowing it to exploit both estimated state and uncertainty. Across four interaction-identifiable manipulation tasks, SSB-Flow outperforms privileged-state distillation baselines and recurrent model-free RL, with the largest gains on tasks requiring inference over higher-dimensional hidden state. These results demonstrate that supervising belief-state inference can make privileged simulation information useful for learning information-gathering manipulation policies.

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